

Supplemental information

Electrochemical sensing of cell viability:

A scalable platform from 2D monolayers

to 3D tissue models

Ignacio Risueño, Sandra Navarro, Maarten van Os, Pablo Acedo, Leticia Valencia, and Diego Velasco

SUPPLEMENTARY MATERIAL

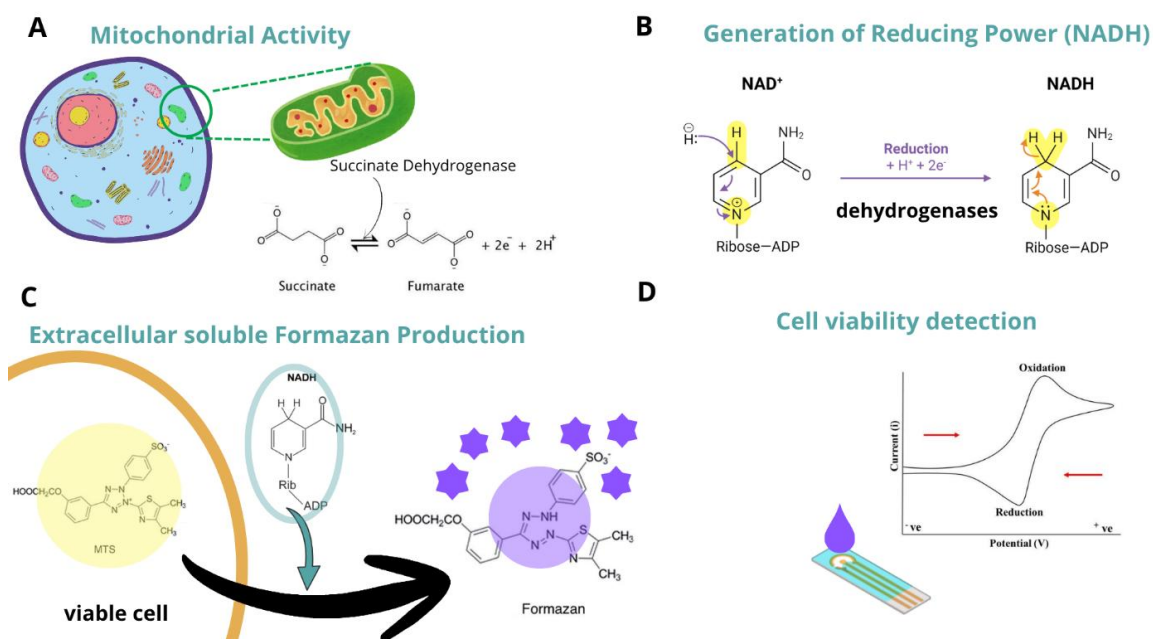


Figure S1. Schematic representation of the electrochemical MTS cell viability detection mechanism.

A) Biological Context: Metabolically active cells utilize mitochondrial enzymes, specifically Succinate Dehydrogenase, as part of the respiratory chain. This activity serves as a direct indicator of cellular health and functional state. **B) Molecular Mechanism:** Mitochondrial dehydrogenases facilitate the reduction of NAD⁺ to NADH, generating the intracellular reducing power required for the assay. **C) Chemical Reaction:** The exogenously added MTS tetrazolium salt enters the cellular environment where it is reduced by NADH to form a highly soluble formazan product. Unlike other tetrazolium salts, the MTS-formazan is readily released into the extracellular culture medium, avoiding cell lysis. **D) Electrochemical Detection:** The soluble formazan is quantified using Screen-Printed Carbon Electrodes (SPCE). The cathodic peak current measured at approximately -0.2 V through Cyclic Voltammetry (CV) is proportional to the concentration of formazan produced, and thus, to the number of viable cells present in the sample. Illustration created using Canva professional design software (Canva Pty Ltd, Australia).

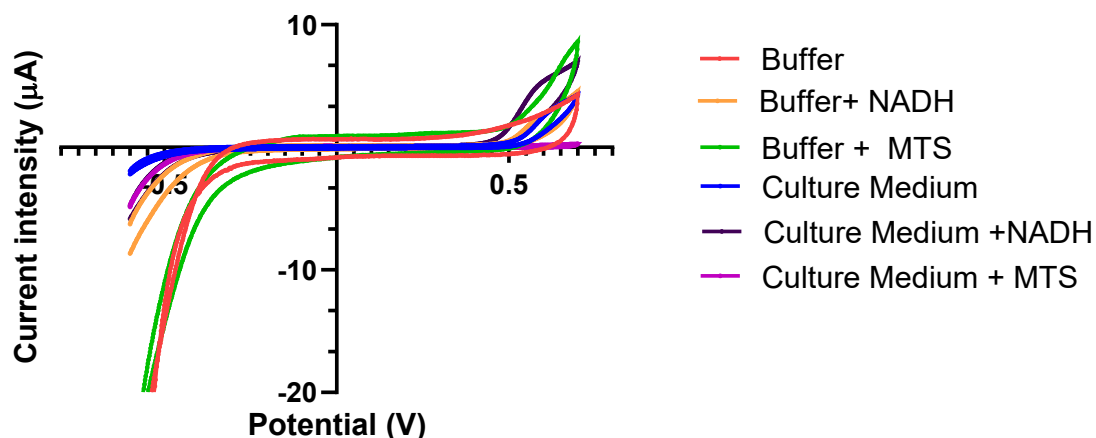


Figure S2. Electrochemical control and selectivity of the MTS/formazan assay.

Cyclic voltammetry scans were performed in different conditions to verify the origin of the electrochemical signal. The absence of redox peaks in these controls confirms that the individual components do not interfere with the detection window. The characteristic peak is only observed in the presence of both MTS and NADH, confirming that the sensor specifically quantifies the reduced MTS product (formazan).

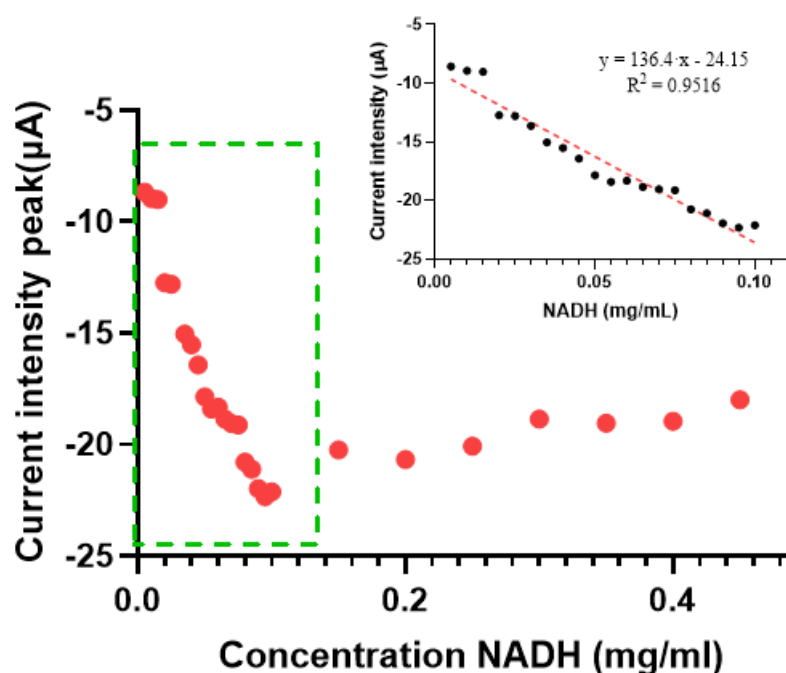


Figure S3. Calibration curve and sensitivity analysis using Cyclic Voltammetry (CV).

Main plot shows the absolute cathodic peak current ($|I_p|$) as a function of NADH concentration over the full experimental range. Inset: Linear regression analysis of the low-concentration regime ($R^2 = 0.9516$), used for the analytical validation of the sensor. The Limit of Detection (LOD) was calculated as $3.3 \sigma / S$, where σ represents the standard deviation of the residuals ($Sy.x = 0.2292$) and S is the slope of the linear fit. The resulting LOD of 0.0047 mg/mL demonstrates the platform's capacity to quantify subtle metabolic changes.

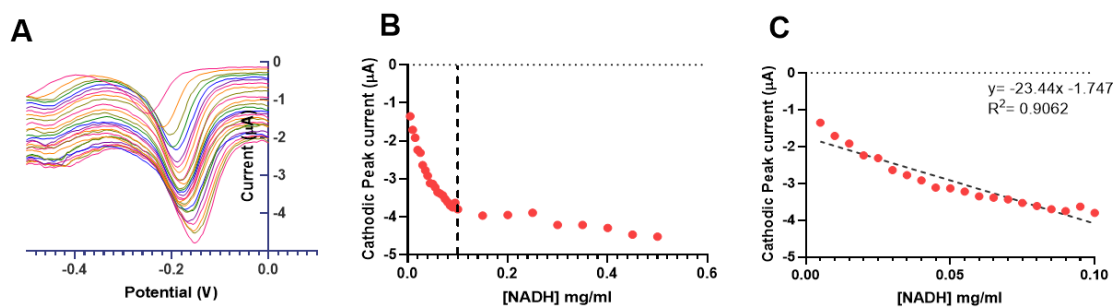


Figure S4. Analytical validation of MTS reduction using Differential Pulse Voltammetry (DPV).

A) DPV curves for different concentrations of reduced MTS (formazan) generated by NADH. **B)** Correlation between the cathodic peak current and NADH concentration. **C)** Linear range analysis at low concentrations, showing the high sensitivity of the SPCE platform for quantitative metabolic monitoring.

Table S1. Comparative analysis of analytical performance and operational features between conventional MTS colorimetry and the proposed electrochemical sensing platform.

Feature	Spectrophotometry (MTS/MTT)	Proposed Electrochemical Method
Optical Interferences	High (Turbidity, pigments)	None (Redox specific)
Path-length dependency	Yes (Limits sensitivity in small volumes)	No (Surface area dependent)
Equipment	Large, expensive plate readers	Portable, low-cost potentiostat
Tissue Preservation	Often requires destructive steps	Fully Non-Destructive
Scalability	High (96/384-well plates)	High (Multichannel/96-well SPCE)

Note: The comparison highlights the advantages of the electrochemical method in 3D tissue models (RHE). Optical interference refers to the impact of sample turbidity or endogenous pigments on signal accuracy. Path-length dependency indicates whether the sensitivity is limited by the physical volume/depth of the sample (Beer-Lambert law)¹. Tissue preservation assesses the capability to perform secondary analyses (e.g., histology or cytokine profiling) on the same biological insert post-measurement². Cost and portability considerations are based on the requirements for a standard microplate reader versus a miniaturized, portable potentiostat with disposable screen-printed electrodes (SPCEs)³.

Table S2. Concentrations and specifications of chemical stressors and control substances used in viability assays.

Chemical	OECD concentration	Classification
Isopropanol	100% (Pure)	No classification
Potassium hydroxide (KOH)	5% aqueous or 8N	Corrosive (Category 2)
1-decanol	100% (Neat)	Irritant/Corrosive (Category 2)
1-bromo-4-chlorobutane	100% (Neat)	Non irritant/Non corrosive
SDS (Sodium Dodecyl Sulfate)	5% aqueous	Positive Control
PBS	Sterile liquid	Negative Control

Note: All chemical concentrations and exposure protocols were selected in accordance with OECD Test Guidelines 431 (Skin Corrosion) and 439 (Skin Irritation). Substances labeled as "Neat" were applied at 100% analytical purity to simulate direct topical exposure on the Reconstructed Human Epidermis (RhE) surface. Aqueous solutions (KOH and SDS) were prepared as weight/volume (w/v) percentages in deionized water. PBS and 5% SDS served as validated negative and positive controls, respectively, ensuring the physiological and cytotoxic benchmarks of the experimental setup.

1. Pour SRS, Calabria D, Emami Amin A, et al. Electrochemical vs. Optical Biosensors for Point-of-Care Applications: A Critical Review. *Chemosensors*. 2023;11(10):546. doi:10.3390/chemosensors11100546
2. Asphahani F, Zhang M. Cellular impedance biosensors for drug screening and toxin detection. *The Analyst*. 2007;132(9):835. doi:10.1039/b704513a
3. Niles AL, Moravec RA, Worzella TJ, Evans NJ, Riss TL. High-Throughput Screening Assays for the Assessment of Cytotoxicity. In: Steinberg P, ed. *High-Throughput Screening Methods in Toxicity Testing*. 1st ed. Wiley; 2013:107-127. doi:10.1002/9781118538203.ch5